

Module 3 - Data Centre Design

Questions:

1. Discuss the characteristic of an outstanding design.[CO3]
2. What are the compulsory local building codes for the datacentre? Explain.[CO3]
3. Describe Aisle structure in data center and how they are used for cooling the data center.
4. Discuss the advantage and disadvantage of raised floor over no raised floor.[CO3]
5. Write a detailed note on cable trays. [CO3]
6. What are the advantages and disadvantage raised floor over the drop ceiling infrastructure of datacentre?Discuss.[CO3]
7. Explain Vandalism in Data center in details.[CO3]

Data center design best practices

There is no single way to design a data center, and countless designs exist that cater to the unique needs of each business. But the following strategies can [produce a data center design with superior efficiency and sustainability](#):

Measure power efficiency. Data center operators can't manage what they don't measure. Use metrics such as PUE to oversee the efficiency of the data center. PUE should be a continuous measurement taken at frequent intervals, year-round, as seasons and weather can affect power usage.

Consider creative space designs. Rather than a traditional, large open space for IT gear, consider a more segregated or compartmentalized data center space. This can include areas that can be left empty if unneeded, used to host different types of gear with similar power/cooling demands or other approaches to reduce power and cooling demands.

Revisit airflow. Cooling is essential to safe operation of IT infrastructure, but airflow must be managed and optimized. This can include limiting hot air/cold air mixing, using hot aisle/cold aisle containment schemes and even using blanking plates to cover unused rack openings, which prevents cooled air from flowing to places that don't cool any gear.

Raise the temperature. The colder a server room is, the more power-hungry and expensive it is. Rather than keeping the server room colder, evaluate the effect of actually raising the temperature. For example, rather than running a cold aisle at 68 to 72 degrees Fahrenheit, consider running the cold aisle at 78 to 80 degrees Fahrenheit. Most IT gear can tolerate elevated temperatures in this way, so long as the gear operates at a steady-state temperature.

Try alternative cooling. An HVAC system might be standard for data centers, but consider ways to reduce or eliminate dependence on conventional HVAC. For example, data centers in cooler climates might reduce HVAC use and introduce cooler outside air -- termed *free cooling* -

- into the facility. Similarly, HVAC can be supplemented or replaced by water-cooled chillers -- i.e., *economizers* -- or other heat-exchange technologies that use far less energy.

Improve power distribution. Data center power efficiency is often lost due to [inefficiency in power-handling](#) and distribution devices, such as equipment transformers, [PDUs](#) and UPS gear. Use high-efficiency power distribution gear and minimize gear -- both of which result in fewer steps -- changes in voltage and current -- and opportunities for power loss.

Data center design challenges

hough there is no single uniform formula for data center design and construction, there are numerous perpetual challenges faced by data center designers and operators. Below are several broad considerations and challenges:

Scalability. A data center is a long-term installation that can remain in service for decades. However, data centers operating today might differ significantly from data centers one or two decades from now. Designers must consider ways of handling today's workloads and services, while also considering how those resources should scale well into the future. The challenge is in providing room for growth in space, power and cooling while mitigating the costs of such capacity until it's needed.

Flexibility. A data center is a bit like a heavy manufacturing floor: Equipment is put into place but can be impossible to move and change as demands evolve. An inability to move gear and shift aisles can prevent businesses from adapting and changing to meet new business demands. The challenge is meeting the needs for change without downtime or costly and time-consuming redesign.

Resilience. A business relies on its data center. If the data center doesn't work, the business doesn't work. Power disruptions, network disruptions, environmental catastrophes and even hacking and other acts of malice can take down a data center. Designers face the challenge of understanding the most prevalent threats and designing appropriate resilience to meet those threats.

Change. New computing technologies and new requirements are always being developed and introduced. Data center designers must consider how to adapt and incorporate often-unforeseeable changes without the need to fundamentally redesign the IT infrastructure to accommodate each change.

Data Center Structure

Structural engineering is responsible for designing and constructing the physical structure of the data centre. This includes the framing and foundation, flooring, walls, and roof. The specific challenges for data centres are providing adequate support for the weight of the equipment, designing fire suppression systems and creating an efficient layout to reduce energy costs.

What is a raised floor?

A raised floor is a data center construction model in which a slightly higher floor is constructed above the building's original concrete slab floor, leaving the open space between the two floors for wiring or cooling infrastructure.

These types of floors are subject to local building codes and national and industry regulatory standards. They should resist wear and abrasion and contain conductive materials with antistatic properties to remove the buildup of damaging static electricity from the environment. The American Society of Heating, Refrigerating and Air-Conditioning Engineers provides structural guidelines for data center raised floors that house HVAC mechanisms.

Types of raised floors explained

There are two main types of raised floor construction: traditional and low profile.

Traditional. These types of floors are constructed using pedestals and floor panels and are typically higher than 6 inches from the concrete floor, also called the *subfloor*. A post stands on the subfloor, below the raised floor. It supports the joint between panels that constitute the raised floor. The traditional model creates ample room for underfloor air distribution coming from the [computer room air handler](#) or any electronic infrastructure that needs to be there. The floor panels are made of varying mixtures of steel, aluminum, cement and wood core.

Low profile. This type of raised floor has a floor height of less than 6 inches from the subfloor. Low-profile raised access floor systems are simpler to install because they require less heavy-duty supportive material. Low-profile floors are suitable for [cable management](#), which can be found in organizations that must run cables or wiring bet

ween the raised floor and the subfloor. They provide easy access to the cables and other components below.

Benefits of raised floors in data centers

Some benefits of using a raised floor model in the data center include the following:

- 1.Conserved materials.** With a raised floor, a data center doesn't require new walls, rooms or structures to house cooling equipment.
- 2.Cost-effectiveness.** Raised floors cool the data center using a minimal number of fans and other HVAC equipment. They also [conserve energy use](#).
- 3.Modular design.** Raised floor systems can often be easily rearranged. Tiles can be added or removed as required. If servers need to be moved, tiles can be rearranged to accommodate the rerouting of cool air. If a data center requires more fans, cabling can be stored under the floor and fans can be stored overhead, separate from the cabling.
- 4.Material flexibility.** Different types of raised floor tile materials and flooring finishes offer various benefits. Certain materials may prevent magnetic or [electrostatic field](#) disruption, as well as providing better traction, safety factors and durability.

Best practices for raised floors in the data center

Done properly, raised flooring products can protect a data center from damage or catastrophic failure while also achieving energy [efficiency goals](#).

One important goal of raised floor implementations in the data center is to reduce the intake air temperature of IT equipment and servers. There are several best practices involved with doing this, including the following:

1.Initial design. It is always easier -- and saves time and money -- to start with a hot and cold aisle design, rather than retrofit an existing data center's architecture later with that type of design.

2.Rack load densities. Architects need to consider the server rack load density to determine how much cooling each unit will need. For example, a higher-density server rack usually creates more heat. The size of the racks will also impact the panels needed to support them to avoid damage.

3.Panel choice. Raised floor vendors offer perforated floor tiles that let different amounts of cool air through. For example, a 25% floor panel consist of approximately 25% open space to let cool air through. The weight of the panel material should also be considered. Some materials may be heavier, thus making flooring installation or rearrangement more difficult. Materials may also vary in their load ratings and the maximum weights they will accommodate.

4.Diffusers. Underfloor [baffles](#) and diffusers can also be used to increase efficiency and guide cool air to server racks.

5.Monitoring. To ensure that racks are cooling and raised floors are working as intended, data center operators should consistently [monitor](#) temperature at the rack level. Alarm thresholds can be set to prevent a loss of cooling to the servers. Monitoring humidity and air velocity are also good practices.

Data Centre Structures

The key infrastructural elements of a data centre structure are:

- Floor (hard floor or raised floor)
- Aisles
- Ramp

These elements make up the essential framework of a data centre. They organise the structure for the HVAC, network connections, servers and storage.

No-raised or Raised Floor

A data centre floor can either be no-raised, also called no-raised floor or be raised. The decision

for a no-raised or raised floor is based on the type of equipment that will be used in the data centre.

No-raised Floor:

This type of floor is usually used in ISP and co-location data centres. Such data centres would have cabinets and cages constructed for their various customers. A co-location is a data centre facility in which a business can rent space for servers and other computing hardware.

The hard floor does not have any plenum space for the cables or air circulation. Many data centres are favouring this type of flooring due to the following advantages:

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It is suitable for large cage construction. The wire-fenced cages can be constructed from the floor to ceiling. Each cage can be leased to a customer.

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It provides additional security. No one can remove a tile, crawl below the floor and get unauthorized access to the equipment.

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Cold air circulates evenly throughout the datacentre, unlike a raised floor structure. With raised floors, the sections close to HVAC may have too many perforated tiles, thus decreasing the cold air circulation to other area

It provides ease of installation and simplicity in design as all the cables and HVAC ducts are located above the cabinets and cages.

Raised floor:

In a raised floor, there is about two feet of plenum space below the tiles. The space is used for the:

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Cabling network

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Power distribution

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Air conditioning

Raised flooring requires more focus because it needs to be strong enough to support the load of heavy equipment. Such floors may need to be reinforced with additional structural support to enable them to bear the weight of the equipment.

There are some advantages of using raised floors in a data centre:

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It enables the cables to be neatly laid out along the wireways and cable trays in the subfloor plenum as shown in Figure. This becomes advantageous for ISP and co-location data centres.

It removes the issues of overhead water pipe leaking. In case of raised floor, the water can be distributed from underfloor, removing the issue of condensation and overhead pipe leaks. It also makes the data centre look a bit less cluttered.

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It does not necessitate the use of specialised contractors to move the air vent tiles for achieving the required temperature. Such modifications become difficult if the air conditioning is using overhead ductwork.

Aisles

An aisle is the space allotted between two rows of racks in a data centre. Figure shows the aisle and open corner space between rows of server cabinets.

While planning the rack layout, it is important to have wide aisles and open space around corners

and walls. The open space is required to move racks and large equipment. Also, the space should be wide enough to move the forklift, which is used to carry large equipment.

If the aisle space is narrow, it would become difficult to move out a broken rack and move in a new one.

The rack layout should also factor in the length of the racks placed side by side. Too many racks lined up in a row would mean more time to navigate from one aisle to another. Gaps should be added after a set of racks so that valuable time and energy is saved trying to get from one aisle to another

Ramp

A ramp is a sloping surface that joins two uneven levels of a structure. A ramp is mandatory when using a raised floor to bridge the gap between the raised floor and floors that are at a different level.

e.g.:

A ramp can be built at the entrance to move the equipment in and out of the structure.

In fact, a ramp is a practical solution to move heavy equipment in or out of a data centre.

The ramp should be strong enough to support the weight of the:

- Equipment itself
- Device for lifting the equipment (such as pallet jacks)
- People handling the equipment and lifting device

The ramp slope should not be too steep. A ramp slope is considered steep if it has a gradient of 1 in 10. This indicates that there is a vertical rise of 1 inch for a horizontal length of 10 inches. The ramp slope should be gradual.

e.g.:

A gradient of 1 in 20 is considered as gradual. So, a 2 foot rise in length would require a horizontal length of 40 feet.

The ramp should ideally be of 80 feet in length and 10 feet in width. It should be wide enough to move large servers and cabinets. It should also be strong enough to carry heavy loads (up to 4000 lbs).

Compulsory Local Building Codes

While building any structure, the architect should plan the design as per the local building codes. Thus, an immense structure, for example, a server farm needs to hold fast to the code or set of configuration particulars that the neighbourhood city authorities characterise

The code specification may specify the minimum ceiling heights of the data centre structure.

Adhering to the building codes is important from the insurance point of view as well. Insurance companies always require a data centre to meet the design specifications, so that any damage to the structure in the event of a natural disaster can be reduced.

In some countries, there would be two levels of approval required from the city authorities:

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Level 1:

The authorities approve the design on paper

Level 2:

The authorities visit and inspect the constructed facility to verify if the structure complies with the approved design plan.

It is important to have an understanding of the codes to prevent any last-minute changes due to non-adherence. Such changes can prove to be very expensive if it relates to the walls or flooring.

Guidelines for Planning a Data Centre

Here are some guidelines that can be followed while designing a data centre:

There should not be any building exterior doors and outside facing windows. Such doors and windows increase the risk of break-in. It may also become difficult to control the data centre temperature. If the structure has such doors or windows, then they should be insulated and enclosed with drywall board.

The data centre perimeter walls should be of full height and hard wall construction.

Do not

build any thin, transparent walls to the interior.

While calculating the required floor space, consider:

- Infrastructure elements (such as HVAC, cables)
- Equipment (such as servers, network equipment)
- Aisle and open space between racks and floor corners
- The data centre floor should ideally be on the ground or the first floor. In case of higher floors, consider the structural strength of the building. Check if the building and the floor can take the weight of the equipment and infrastructure.

The entrance door to the data centre should be wide. It should be at least 8 feet high and 8

feet wide, so that large equipment can be moved easily.

- In case of raised floor, the dimensions of a floor tile should be 600 x 600. The tile material

should be seismic braced in case of earthquake prone areas.

- Use perforated tiles as exit points in case of under the floor air conditioning.

- Plan for a storage area to store equipment temporarily. This storage area should be approximately 15% of the total area of the data centre.

- There should be an unobstructed pathway between the equipment room, storage room and

building exterior. A minimum clearance of 8 feet x 4 feet should be given to all the corridors, doorways and elevators in this pathway.

- The ramp should be constructed using the same material as the raised floor. Cover the ramp

with slip-resistant treads, so that it is safe and easy to move the equipment over the ramp.

- The ramp should meet the same weight loading specifications as that of the raised floor.

Many architects make ramps by pouring concrete to ensure that it can bear heavy weights.

- The raised floor, ramps and subfloor should be strong enough to support large fully loaded

server cabinets that can weigh up to 900 kg.

- The floor layout should be planned in such a way that around 65% of the area is occupied by the racks and stand-alone equipment.

Introduction to Data Centre Design- Part II

A data centre can be constructed on a piece of land or within an existing structure. Once the basic

structure is constructed, the next step is to design the internal infrastructure of the data centre.

Floor is one of the most basic infrastructures of the data centre, which needs to be constructed

properly. Since the entire equipment stands on the floor, it should be designed very carefully to

avoid any retrofitting cost.

Based on the needs of the data centre, the floor can either be a hard floor or raised floor.

Raised

floor has its own benefits related to fitting of electrical wireways and air conditioning.

Raised Floor Design and Deployment

As you already know, once the raised floor is constructed, it is going to stay. Retrofitting a floor

is a tedious affair. So, plan the floor infrastructure carefully.

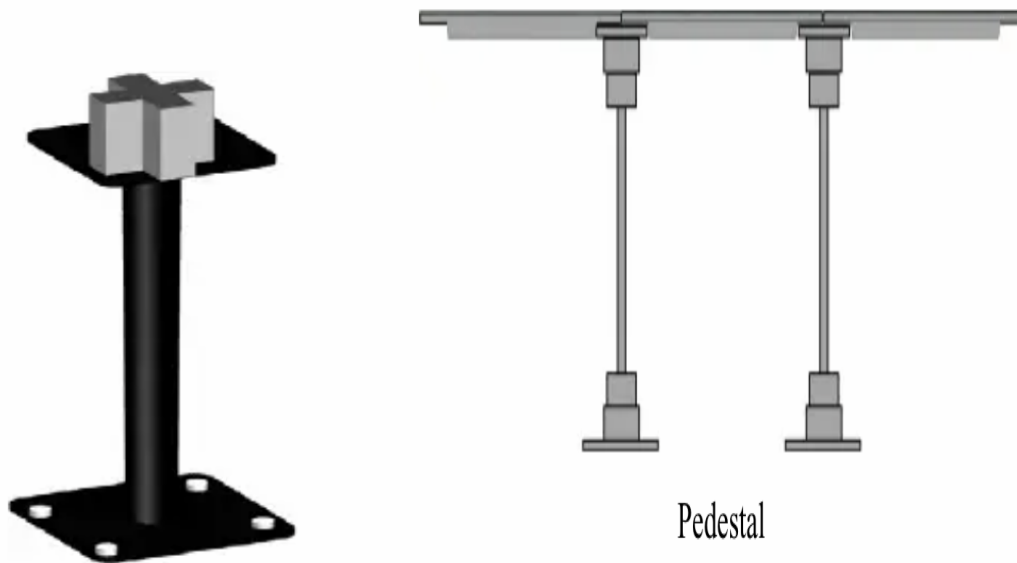
Figure shows the floor grid, which supports the raised floor on which the equipment is kept



Fig. Floor grid supporting the raised floor with equipment

As seen in Figure, a raised floor is constructed on a ground, which has a framework of:

- Pedestals
- Pedestal head
- Stringers•



Pedestal Head

Pedestal

Pedestals:

The pedestals support the floor tiles. Generally, the floor tiles are 2 feet × 2 feet in size. So, the pedestals are arranged such that they support each floor tile.

Pedestal

•Pedestal head:

It is fitted on top of the pedestal. The pedestal head supports the stringers and the floor tiles.

Stringers:

These are steel or aluminium channels that are fastened to the pedestal head.

Stringers provide lateral stability to the raised floor.

In case of an earthquake prone area, the floor grid system should be seismic braced so that it can withstand the tremors.

The raised floor of a data centre provides:

- A place on which the equipment is kept. Therefore, the floor tiles must be able to endure the weight of the equipment.
- Grounding for the equipment.
- Air circulation of cold air. The cold air from the Heating, Ventilation and Air Conditioning (HVAC) units within the subfloor enclosure is directed up to the data centre through the penetrated tiles to cool the equipment.
- An area to lay out the network cables, cables for equipment and power outlets. When the cables are kept under the floor area, the data centre looks less filled. In this arrangement, cables remain safe from damages caused by accidental step over.

Plenum

Plenum is the space between the data centre subfloor and floor tiles. The height of the plenum is usually between 1.5 feet to 2 feet. This space is determined by the capacity of the HVAC as well.

The HVAC should be able to pressurise the plenum so that the cold air could circulate.

The plenum space is occupied by the floor grid system consisting of pedestals and stringers.

Floor Tiles

The floor panels or tiles rest on the pedestals and stringers. It forms a supporting base for the data centre equipment. The tiles are usually in the form of a square with dimension of 2 feet x 2 feet.

The material of the tiles is important in determining the weight that it can support. Tiles can be made of concrete, compressed wood or cast aluminium. Usually, cast aluminium is used as material for the tiles. Such tiles can support a weight of up to 680 kg.

The weight bearing capacity of concrete or compressed wood tiles is less (up to 225 kg) as compared to cast aluminium tiles. If the correct tiles are not used, then they may crack due to

pressure of the equipment and may cause damage to the expensive equipment.

Tiles can be of different types depending on its function:

Solid tiles:

Do not have holes in them and are used to support the equipment. These tiles redirect the air flow and thus help in preserving the pressure in the plenum.

Perforated tiles:

They have holes in them to allow cold air to enter from the plenum. These tiles are used right next to equipment racks and below the heavy equipment that require its bottom to be cooled. This placement allows the cold air directly to equipment rack

Based on the shape and size of the data centre floor, sometimes, the tiles have to be cut so that it can fit on the edges of the floor and around columns. In such cases, the cut edges of the tiles should be capped with protective trims. This structure will avoid particles to enter the airstream.

Equipment Weight and Tile Strength

The size and weight of the equipment has a great impact on the design of the floor tiles as large equipment can stress the floor structure. To determine whether a tile would be able to bear the equipment load, the current as well as the expected future load should be considered. The initial design should not spread out the equipment on all of the available floor space.

Therefore, the architect should know the approximate weight of the equipment and load bearing capacity of the tiles before designing the subfloor grid system.

There are two types of loads that are calculated to understand the load bearing capacity of the tiles. They are:

- Point load
- Static load
- Point load:

Most of the equipment or racks in a data centre rest on four rollers, casters or feet. Point load is the load on any one of these four feet.

e.g.

A large server weighs 2,400 pounds. Its point load would be $2,400 / 4 = 600$ pounds.

This point load is for each of its feet. If the feet of this server rest on one square inch, then the tile should be able to bear 600 pounds on one square inch without deflection of more than 1 mm or 2 mm.

- Static load:

This load refers to the sum of all the point loads on the tile.

e.g.:

Consider, there are two racks with point loads of 600 pounds per feet. Now, if each of the racks has one foot on a particular tile, then that tile will have a total load of $(600 + 600) = 1,200$ pounds. In this case, the tile should be rated for at least 1,200 pounds of static load.

This calculation is not only for solid tiles, but also for perforated tiles. Generally, perforation weakens a tile. Therefore, when equipment is moved along the aisle, the perforated tiles would experience the static load temporarily. Hence, even perforated tiles should have adequate static load rating.